5

Introduction to Probability and Sample Statistics

Chapter Outline

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Key Concepts

1. Probability
2. Inferential statistics
3. Simple random sampling (with and without replacement)
4. Sampling distribution of the mean
5. Variance and standard error of the mean (sampling error)
6. Confidence intervals (point vs. interval estimation)
7. Central limit theorem

In Chapter 4 we extended our discussion of descriptive statistics. We considered the following three general topics: the normal distribution, standard scores, and skewness and kurtosis. In this chapter we begin to move from descriptive statistics into inferential statistics (in which normally distributed data plays a major role). The two basic topics described in this chapter are (a) probability and (b) sampling and estimation. First, as a brief introduction to probability, we discuss the importance of probability in statistics, define probability in a conceptual and computational sense, and discuss the notion of intuition versus probability. Second, under sampling and estimation, we formally move into inferential statistics by considering the following topics: simple random sampling (and briefly other types of sampling), and estimation of population parameters and sampling distributions. Concepts to be discussed include probability, inferential statistics, simple random sampling (with and without replacement), sampling distribution of the mean, variance and

standard error of the mean (sampling error), confidence intervals (point vs. interval estimation), and central limit theorem. Our objectives are that by the end of this chapter, you will be able to (a) understand the most basic concepts of probability; (b) understand and conduct simple random sampling; and (c) understand, determine, and interpret the results from the estimation of population parameters via a sample.

5.1 Brief Introduction to Probability

The area of probability became important and began to be developed during the Middle Ages (17th and 18th centuries) when royalty and other well-to-do gamblers consulted with mathematicians for advice on games of chance. For example, in poker if you hold two jacks, what are your chances of drawing a third jack? Or in craps, what is the chance of rolling a 7 with two dice? During that time, probability was also used for more practical purposes, such as to help determine life expectancy to underwrite life insurance policies. Considerable development in probability has obviously taken place since that time. In this section, we discuss the importance of probability, provide a definition of probability, and consider the notion of intuition versus probability. Although there is much more to the topic of probability, here we simply discuss those aspects of probability necessary for the remainder of the text. For additional information on probability, take a look at texts by Rudas (2004) or Tijms (2004).

5.1.1 Importance of Probability

Let us first consider why probability is important in statistics. A researcher is out collecting some sample data from a group of individuals (e.g., students, parents, teachers, voters, corporations, animals, etc.). Some descriptive statistics are generated from the sample data. Say the sample mean, $\bar{X}$, is computed for several variables (e.g., number of hours of study time per week, grade point average, confidence in a political candidate, widget sales, animal food consumption). To what extent can we generalize from these sample statistics to their corresponding population parameters? For example, if the mean amount of study time per week for a given sample of graduate students is $\bar{X} = 10$ hours, to what extent are we able to generalize to the population of graduate students on the value of the population mean, μ?

As we see, beginning in this chapter, inferential statistics involve making an inference about population parameters from sample statistics. We would like to know (a) how much uncertainty exists in our sample statistics, as well as (b) how much confidence to place in our sample statistics. These questions can be addressed by assigning a probability value to an inference. As we show beginning in Chapter 6, probability can also be used to make statements about areas under a distribution of scores (e.g., the normal distribution). First, however, we need to provide a definition of probability.

5.1.2 Definition of Probability

In order to more easily define probability, consider a simple example of rolling a six-sided die (as there are dice with different numbers of sides). Each of the six sides, of course, has

probability of tossing 10 consecutive heads, which is rather small (approximately .0010). So when you are gambling at the casino and have lost the last nine games, do not believe that you are guaranteed to win the next game. You can just as easily lose game 10 as you did game 1. The same goes if you have won a number of games. You can just as easily win the next game as you did game 1. To some extent, the casinos count on their customers playing the gambler's fallacy to make a profit.

5.2 Sampling and Estimation

In Chapter 3 we spent some time discussing sample statistics, including the measures of central tendency and dispersion. In this section we expand upon that discussion by defining inferential statistics, describing different types of sampling, and then moving into the implications of such sampling in terms of estimation and sampling distributions.

Consider the situation where we have a population of graduate students. **Population parameters** (which are characteristics of a population) could be determined, such as the population size (N), the population mean (μ), the population variance (σ^2), and the population standard deviation (σ). Through some method of sampling, we then take a sample of students from this population. **Sample statistics**, which are just characteristics of a sample, could be determined, such as the sample size (n), the sample mean ($\bar{X}$), the sample variance (s^2), and the sample standard deviation (s).

How often do we actually ever deal with population data? Except when dealing with very small, well-defined populations, we almost never deal with population data. (There are always exceptions; however, our experience dictates that it is almost always the case that we are working with sample data.) The main reason for this is cost, in terms of time, personnel, and economics. *This means then that we are almost always dealing with sample data*. With descriptive statistics, dealing with sample data is very straightforward, and we only need to make sure we are using the appropriate sample statistic equation. However, what if we want to take a sample statistic and make some generalization about its relevant population parameter? For example, you have computed a sample mean on grade point average (GPA) of $\bar{X} = 3.25$ for a sample of 25 graduate students at State University. You would like to make some generalizations from this sample mean to the population mean (μ) at State University. How do we do this? To what extent can we make such a generalization? How confident are we that this sample mean actually represents the population mean?

This brings us to the field of inferential statistics. We define **inferential statistics** as *statistics that allow us to make an inference or generalization from a sample to the population*. In terms of reasoning, *inductive reasoning* is used to infer from the specific (the sample) to the general (the population). Thus, inferential statistics is the answer to all of our preceding questions about generalizing from sample statistics to population parameters. *How* the sample is derived, however, is important in determining to what extent the statistical results we derive can be inferred from the sample back to the population. Thus, it is important to spend a little time talking about simple random sampling, the only sampling procedure that directly allows generalizations to be made from the sample to the population. Although there are statistical means to correct for non-simple random samples, they are beyond the scope of this textbook. Researchers may wish to refer to references, such as Skinner, Holt, and Smith (1989). In the remainder of this section, and in much of the remainder of this text, we take up the details of inferential statistics for many different procedures.

distribution is nonnormal, then this will have minimal effect on the sampling distribution of the mean except for rather small samples. This can come into play with inferential statistics when the assumption of normality is not satisfied, as we see in later chapters.

5.3 Additional Resources

This chapter is meant to serve as a concise and general introduction to probability, sampling, and related concepts. For readers who want more in-depth and comprehensive coverage, numerous superb references are available to assist in learning more about concepts introduced in this chapter. A number of these have already been cited. Additional resources you may wish to consider include the following:

- Probabilities in the context of everyday examples (Olofsson, 2007).
- A general introduction to probability and statistics (Kinney, 2015).
- An edited work that is a compilation of papers related to teaching and learning probability, valuable both to those teaching probability as well as those learning probability (Batanero & Chernoff, 2018).

Appendix 5.a: Probability That at Least Two Individuals Have the Same Birthday

This probability can be shown by either of the following equations. Note that there are $n = 23$ individuals in the room. One method is as follows:

$$1-\left(\frac{(365)(364)(363)\ldots(365-n+1)}{365^n}\right)=1-\left(\frac{(365)(364)(363)\ldots(343)}{365^{23}}\right)=.507$$

An equivalent method is as follows:

$$1-\left[\left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right)\ldots\left(\frac{365-n+1}{365}\right)\right]=1-\left[\left(\frac{365}{365}\right)\left(\frac{364}{365}\right)\left(\frac{363}{365}\right)\ldots\left(\frac{343}{365}\right)\right]=.507$$

Problems

Conceptual Problems

1. The standard error of the mean is which of the following?

a. Standard deviation of a sample distribution
b. Standard deviation of the population distribution
c. Standard deviation of the sampling distribution of the mean
d. Mean of the sampling distribution of the standard deviation